



05506026

Digital Image Processing

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**MATHEMATICAL TOOLS USED IN
DIGITAL IMAGE PROCESSING**



PIXELS RELATIONSHIPS

Image is a set of pixels

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(1, 0) & f(2, 0) & \dots \\ f(0, 1) & f(1, 1) & f(2, 1) & \dots \\ f(0, 2) & f(1, 2) & f(2, 2) & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Relationships between pixels

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(1, 0) & f(2, 0) \\ f(0, 1) & f(1, 1) & f(2, 1) \dots \\ f(0, 2) & f(1, 2) & f(2, 2) \\ \vdots & & \end{bmatrix}$$

Neighbours of a pixel

- Four horizontal and Vertical neighbours
 - $(x+1, y)$, $(x-1, y)$, $(x, y+1)$, $(x, y-1)$

Relationships between pixels

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(1, 0) & f(2, 0) \\ f(0, 1) & f(1, 1) & f(2, 1) \dots \\ f(0, 2) & f(1, 2) & f(2, 2) \\ \vdots & & \end{bmatrix}$$

Neighbours of a pixel

- Four diagonal neighbours
 - $(x+1, y+1)$, $(x+1, y-1)$, $(x-1, y+1)$, $(x-1, y-1)$

Relationships between pixels

Adjacency

- ▶ Two pixels are adjacent if they are neighbours AND their intensity level V satisfy some criteria
- ▶ For example
 - ▶ $V = \{1\}$
 - ▶ $V = \{0, 1\}$ (binary image)
 - ▶ $V = \{0, 1, 2, \dots, 255\}$ (Grayscale image)

Relationships between pixels

Adjacency

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 - ▶ $V = \{0, 1, 2, \dots, 255\}$ (Grayscale image)

Relationships between pixels

Adjacency

- ▶ 4 – adjacency
 - ▶ Two pixels P and Q which contain values in set V are 4 – adjacency if Q is in the set of $N_4(P)$
 - ▶ For example, $V = \{0, 1\}$

1	1	0
1	1	0
1	0	1

Relationships between pixels

Adjacency

- ▶ 8 – adjacency
 - ▶ Two pixels P and Q which contain values in set V are 8 – adjacency if Q is in the set of $N_8(P)$
 - ▶ For example, $V = \{1, 2\}$

0	1	1
0	2	0
0	0	1

Relationships between pixels

Adjacency

- ▶ m – adjacency
 - ▶ Two pixels P and Q which contain values in set V are m – adjacency if
 - ▶ Q is in $N_4(P)$ OR
 - ▶ Q is in $N_D(P)$ and the set $N_4(P) \cap N_4(Q)$ has no pixels whose values are from V

Relationships between pixels

Adjacency

- ▶ m – adjacency
- ▶ For example, $V = \{1\}$
 - ▶ Q is in $N_4(P)$

0	1	1
0	1	0
0	0	1
 - ▶

0	1	0
0	1	0
0	0	1

Relationships between pixels

Adjacency

- ▶ m – adjacency
- ▶ For example, $V = \{1\}$
 - ▶ Q is in $N_D(P)$ and the set $N_4(P) \cap N_4(Q)$ has no pixels whose values are from V

0	1	1
0	1	0
0	0	1

Relationships between pixels

Paths

- ▶ A path from pixel $P(x, y)$ to pixel $Q(s, t)$ is a sequence of distinct coordinates $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where $P(x, y) = (x_0, y_0)$ and $Q(s, t) = (x_n, y_n)$ and all pixels are neighbours to the one next to them

Relationships between pixels

Connectivity

- ▶ Let S represents subset of pixels in an image
- ▶ Two pixels are connected if there is an existing path between them which the entire pixels in this path are in S

Relationships between pixels

Region

- ▶ Let R be a subset of pixels in an image
- ▶ R is a region of the image if R is a connected set

Relationships between pixels

Region

- ▶ R_i and R_j are adjacent if their union forms a connected set
- ▶ Regions that are not adjacent are said to be **disjoint**

Relationships between pixels

Boundary (or Border or Contour)

- ▷ Boundary of a region R is the set of pixels that are adjacent to points in the complement of R (has at least one background neighbor)

Relationships between pixels

Distance

- ▶ **Euclidean** distance between P and Q is defined as

$$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

1.41	1	1.41
1	0	1
1.41	1	1.41

Relationships between pixels

Distance

- **City-block** distance between P and Q is defined as

$$D_4(p, q) = |x - s| + |y - t|$$

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

Relationships between pixels

Distance

- ▶ **Chessboard** distance between P and Q is defined as

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2



MATHEMATICAL TOOLS USED IN DIGITAL IMAGE PROCESSING

Array vs Matrix Operations

- ▶ Digital Image consists with 2D array of pixels which is integers which represent colour data

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(1, 0) & f(2, 0) \\ f(0, 1) & f(1, 1) & f(2, 1) \dots \\ f(0, 2) & f(1, 2) & f(2, 2) \\ \vdots & & \end{bmatrix}$$



128	30	123
232	123	321
123	77	89
80	255	255

Array vs Matrix Operations

- There are two operations that can be carried out using matrix theory

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(1, 0) & f(2, 0) \\ f(0, 1) & f(1, 1) & f(2, 1) \dots \\ f(0, 2) & f(1, 2) & f(2, 2) \\ \vdots & & \end{bmatrix}$$



128	30	123
232	123	321
123	77	89
80	255	255

Array vs Matrix Operations

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

- ▶ If there are two matrices represent two 2 x 2 pixel images

Array vs Matrix Operations

- ▶ MATRIX PRODUCT of two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Array vs Matrix Operations

- ▶ While ARRAY PRODUCT of two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Array vs Matrix Operations

- ▶ In digital image processing, most of operations (unless stated otherwise) are array operations

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Arithmetic Operations

- ▶ Arithmetic operations between images are array operations which are carried out between corresponding pixel pairs

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$

Arithmetic Operations

- ▶ Adding operation is used a lot to add noise to the image.
- ▶ On the other hand, it can be used to denoise the corrupted images

$$s(x, y) = f(x, y) + g(x, y)$$



Image from: https://www.researchgate.net/figure/Effect-of-adding-Gaussian-noise-to-an-image-a-Original-watermarked-image-b-Gaussian_fig2_229008840

Arithmetic Operations

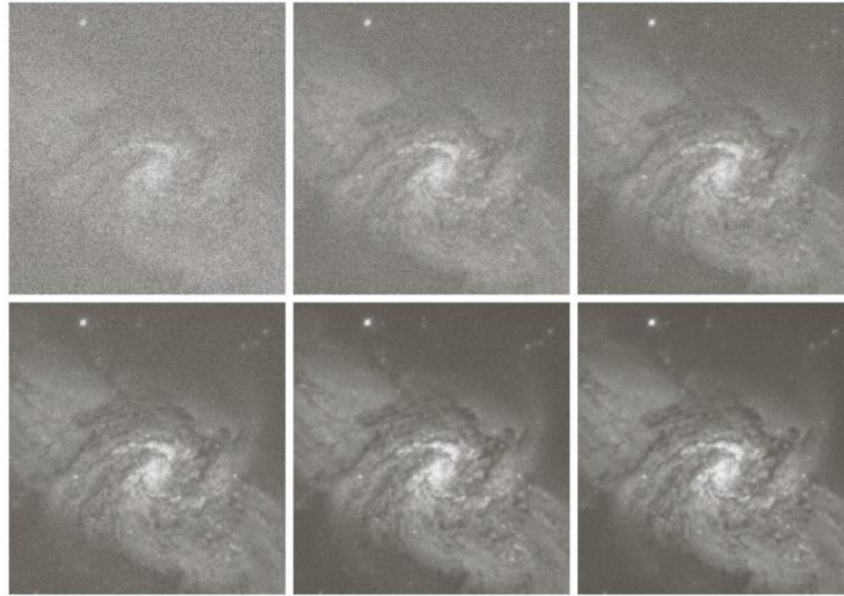


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Arithmetic Operations

- ▶ Subtract operation is used to find the differences between two images

$$d(x, y) = f(x, y) - g(x, y)$$



Arithmetic Operations

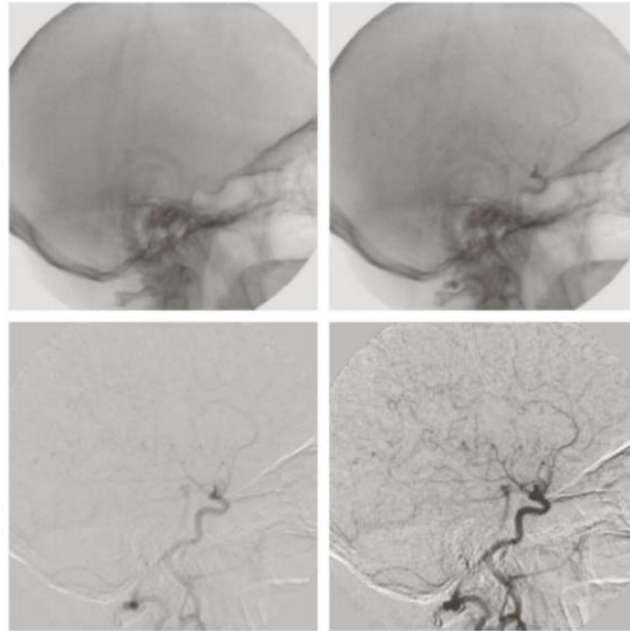


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Arithmetic Operations

- ▶ Multiplication and division can be used for masking the ROI (region of interest)

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$



Arithmetic Operations

- ▶ Another application of multiplication and division operations is shading correction



Linear vs Non-Linear Operations

- ▶ Considering an H operation which give an output image $g(x, y)$ from an input image $f(x, y)$

$$H[f(x, y)] = g(x, y)$$

- ▶ H is a linear operation if

$$\begin{aligned} H[a_i f_i(x, y) + a_j f_j(x, y)] &= a_i H[f_i(x, y)] + a_j H[f_j(x, y)] \\ &= a_i g_i(x, y) + a_j g_j(x, y) \end{aligned}$$

Linear vs Non-Linear Operations

▶ Example

$$f_1 = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix}, f_2 = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix}$$

$$a_1 = 1, a_2 = -1$$

Linear vs Non-Linear Operations

Using SUM operation

LEFT

$$\text{sum} \left[(1) \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right] = \text{sum} \begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} = -15$$

RIGHT

$$(1)\text{sum} \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1)\text{sum} \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} = (0 + 2 + 2 + 3) - (6 + 5 + 4 + 7) = -15$$

Therefore, SUM operation is a linear operation

Linear vs Non-Linear Operations

Using MAX operation

LEFT

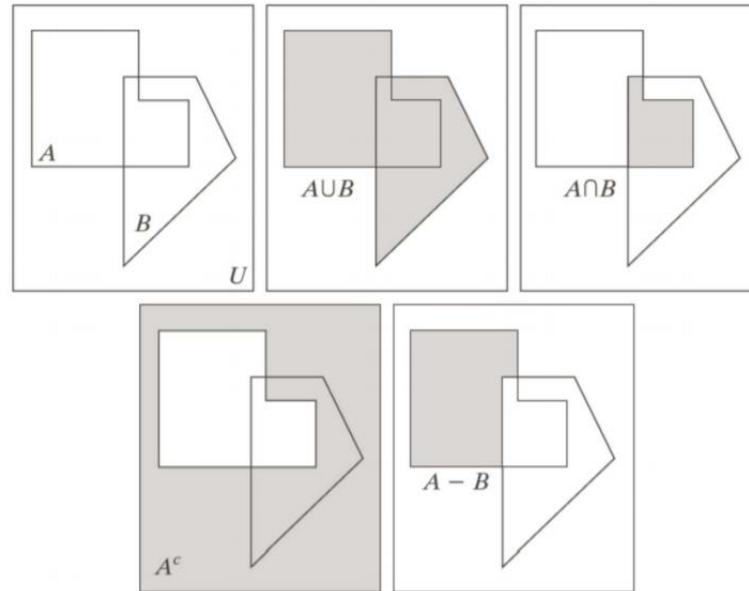
$$\max \left[(1) \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right] = \max \begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} = -2$$

RIGHT

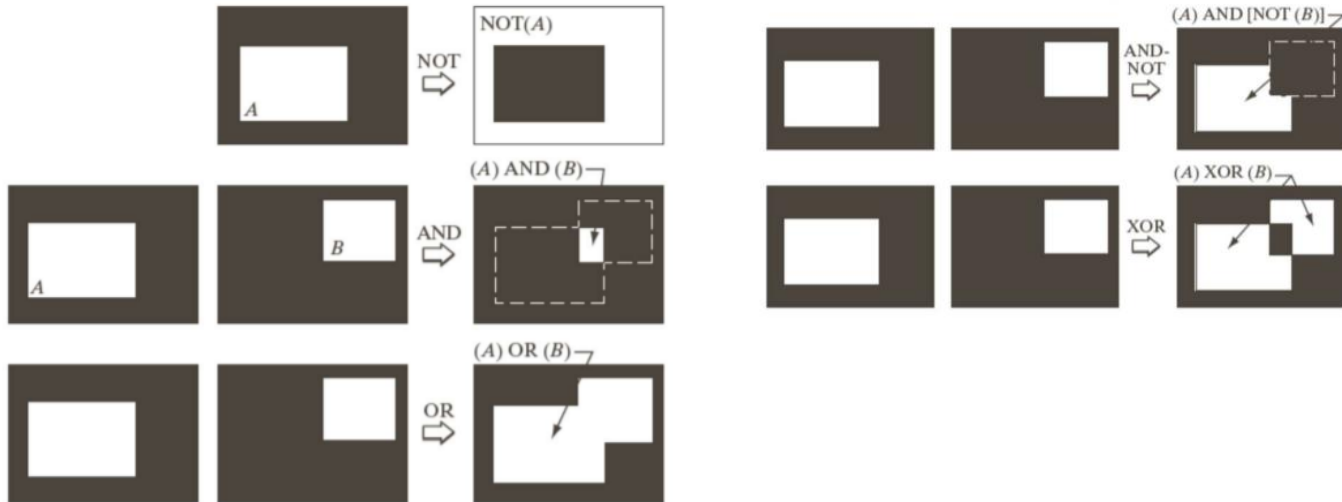
$$(1) \max \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) \max \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} = 3 - 7 = -4$$

Therefore, MAX operation is a non-linear operation

Set and Logical Operations



Set and Logical Operations





End of Week 2